

Leveraging Containers for Deploying Generative AI Applications

Containers offer numerous benefits when deployed for generative AI applications, including portability, scalability, and reliability.



Generative AI is revolutionising various industries by enabling the creation of new content, from text and images to music and code. However, deploying generative AI applications can be complex due to the need for scalable, reliable, and efficient infrastructure. This is where containers, Kubernetes, and Docker come into play. These open source technologies provide a robust framework for deploying and managing generative AI applications, ensuring they run smoothly and efficiently.

Typical enterprise generative AI applications consist of multiple layers (Figure 1).

Application layer: This layer encompasses the user-facing components of the generative application. It includes:

- **User interface:** Web or mobile interfaces through which users interact with the application. For small-scale applications and proof-of-concept use cases, open source frameworks such as Streamlit and Gradio are used widely. For production deployments, web frontends are being used.
- **API gateway:** This is a web frontend interfaced with a backend API. Python FastAPI is a popular choice.
- **Business logic:** Application-specific

logic integrates model predictions with other functionalities. Open source frameworks such as LangChain and LangFlow are widely used to isolate component interactions. They help to create plug-and-play models, eliminating entire code rewrites.

Data layer: AI/ML applications include data processing pipelines for data preparation such as unnecessary noise/data removal, filtering personally identifiable information (PII), and pre-processing such as tokenization. Popular open source vector databases are FAISS (Facebook AI Similarity Search) and Chroma. Enterprises are